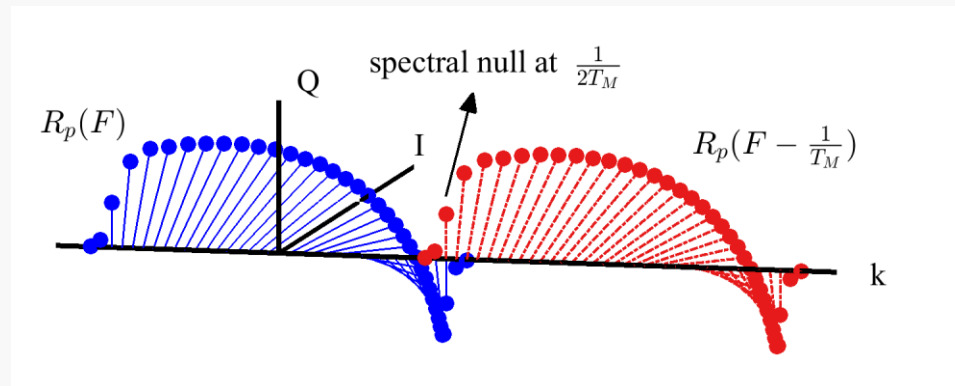


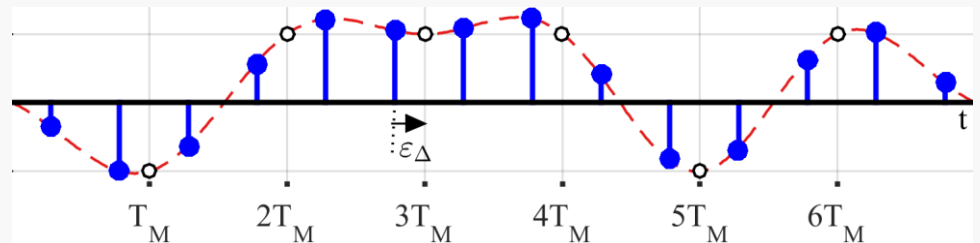
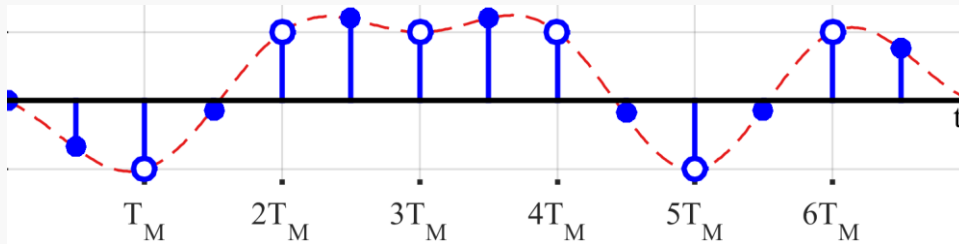
Timing or Clock Synchronization

Qasim Chaudhari



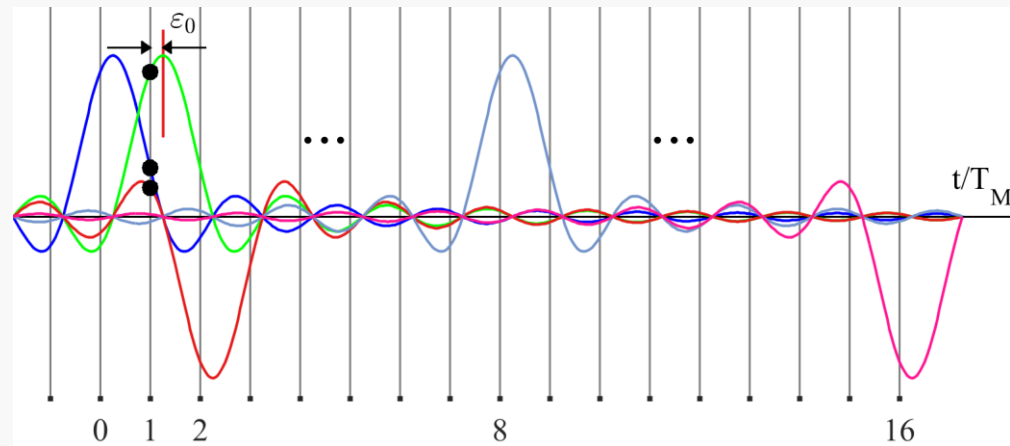
What is Timing Synchronization

- Rx needs to sample the signal every T_M seconds at symbol boundaries
- Results in zero Inter-Symbol Interference (ISI)



Effect of Timing Mismatch

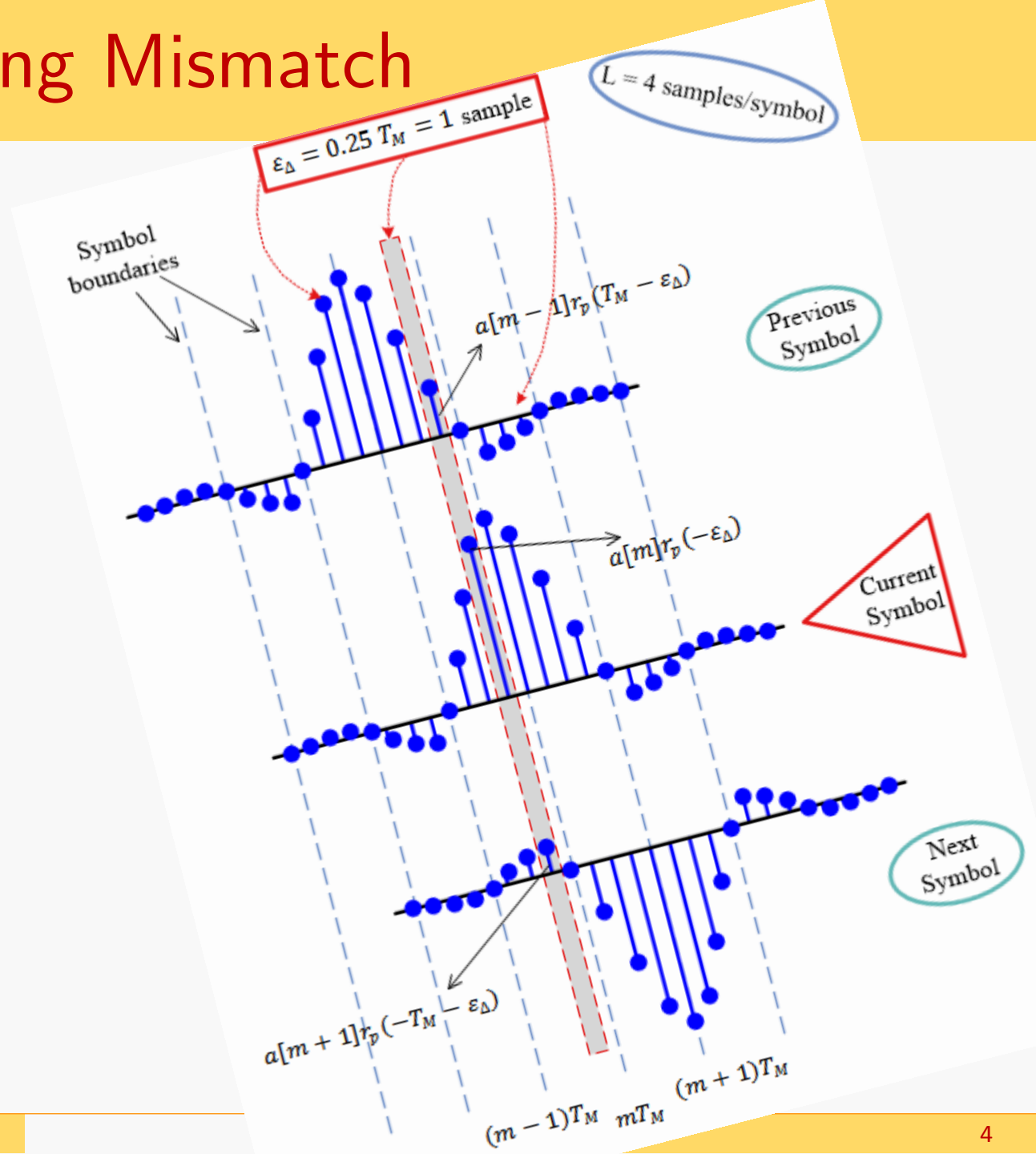
Rx signal is delayed with respect to the Tx signal



A fractional value within a symbol interval T_M

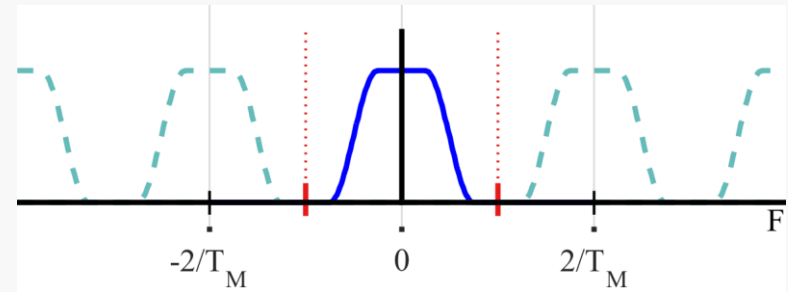
$$-0.5 \leq \varepsilon_0 = \frac{\varepsilon \Delta}{T_M} \leq +0.5$$

Effect of Timing Mismatch

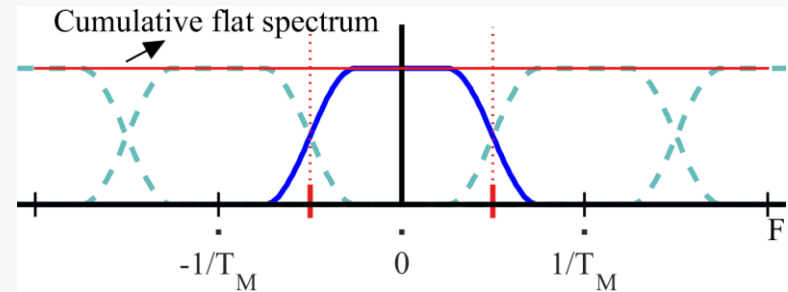


Effect of Timing Mismatch

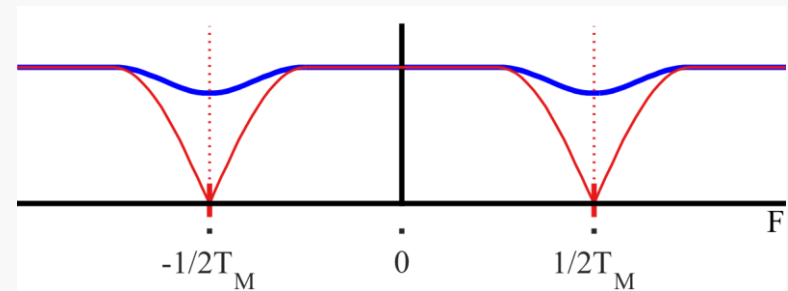
- $F_S = 2$ samples/symbol



- $F_S = 1$ sample/symbol, $\varepsilon_\Delta = 0$



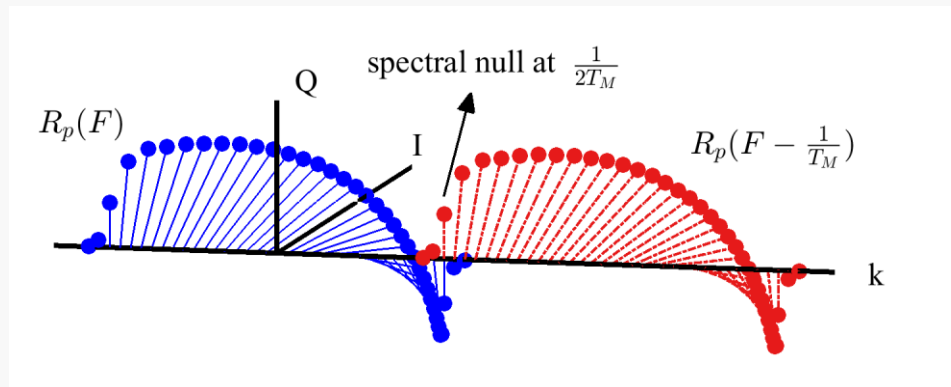
- $F_S = 1$ sample/symbol, $\varepsilon_\Delta \neq 0$



Effect of Timing Mismatch

A 3D view of spectral null mismatch

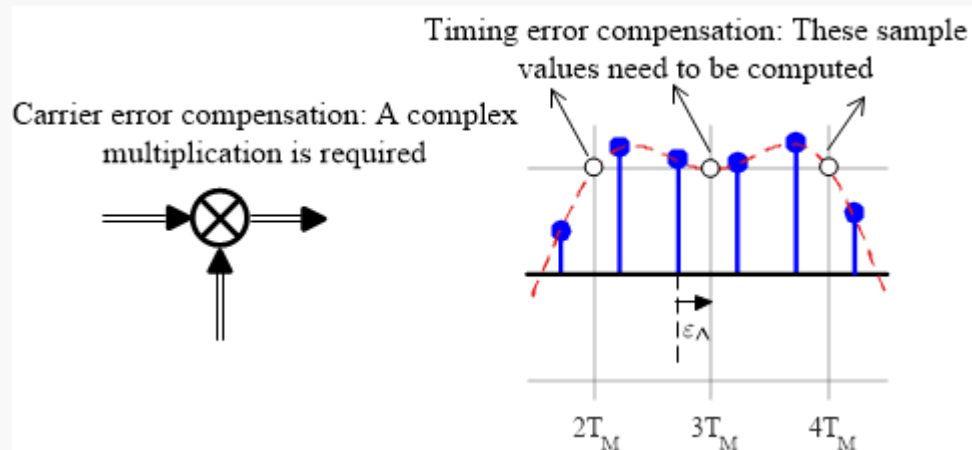
Shift in one domain $\longrightarrow$ Multiplication by a complex sinusoid in the other domain



$$\begin{aligned} s(t) &\longrightarrow S(F) \\ s(t - t_0) &\longrightarrow S(F) \cdot \exp(-j2\pi F t_0) \end{aligned}$$

The Big Picture

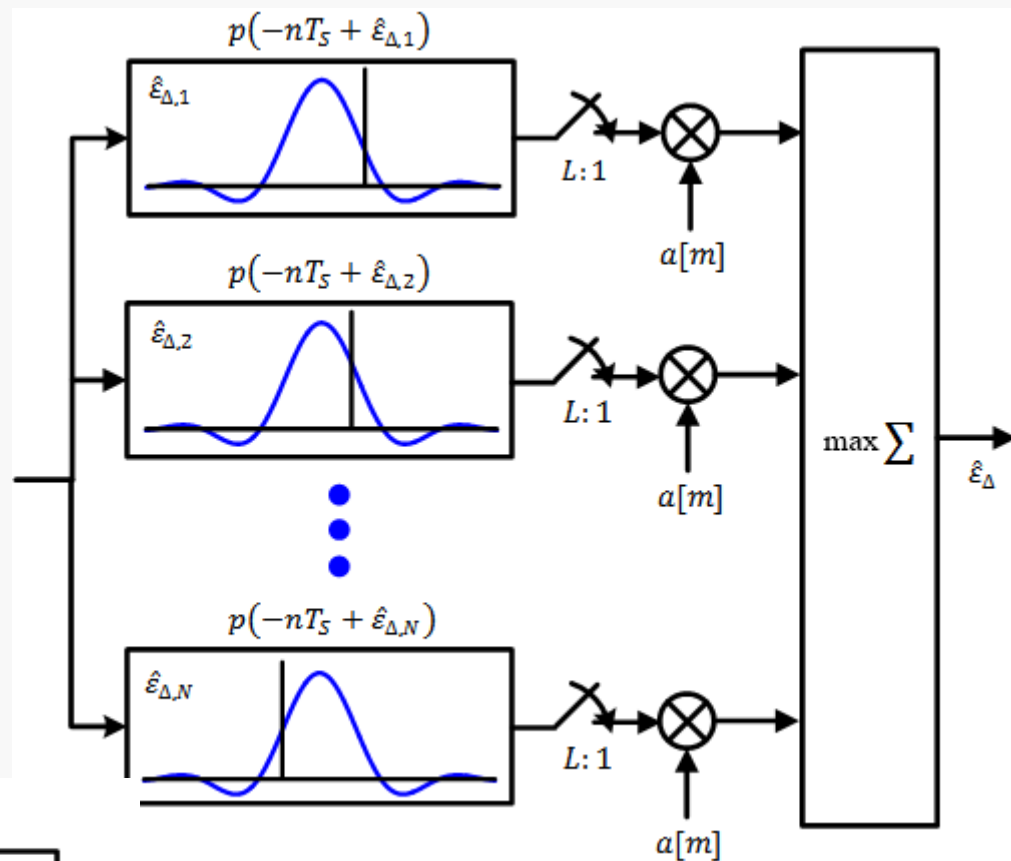
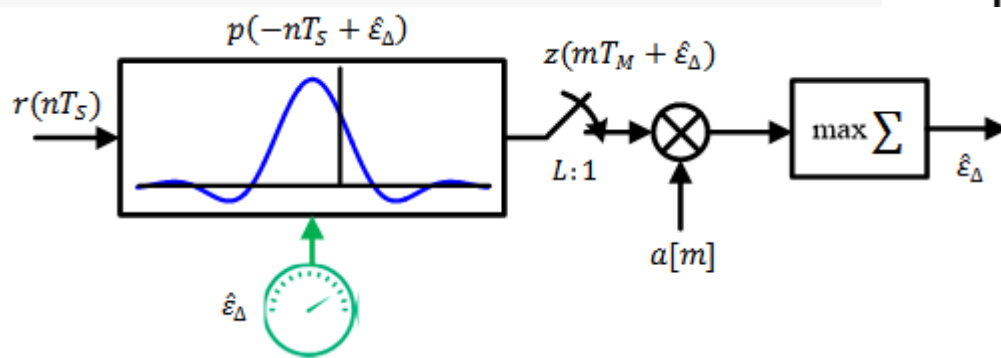
- Data-aided and non-data-aided (or decision-directed) schemes
- Feedforward and feedback techniques are available but the latter is more popular
- How to compensate?
- We will mostly focus on PAM



Brute Force Estimator

Apply the correlation principle

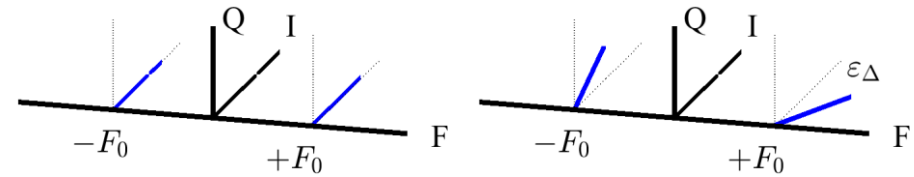
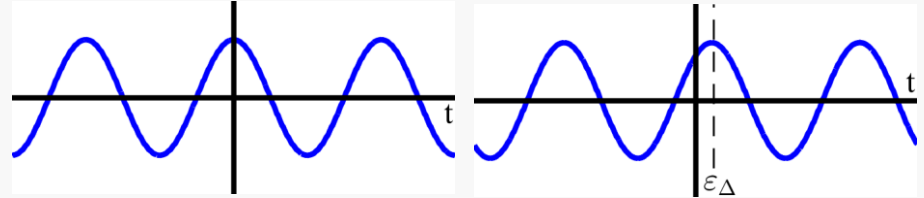
$$\text{corr}[0] = \sum_{m=0}^{N_0-1} a[m] \cdot z(mT_M + \hat{\epsilon}_\Delta)$$



Why Squaring?

For sinusoidal signals

Time shift = Phase shift



Example: shift a cosine wave by $T_0/4$

$$-2\pi F_0 \varepsilon_{\Delta} = -2\pi F_0 \frac{1}{4F_0} = -\frac{\pi}{2}$$

It becomes a sine wave!

$$\sin \theta = \cos(\theta - \pi/2)$$

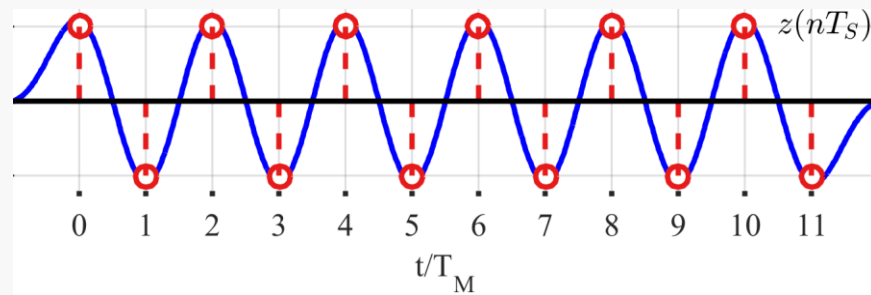
$$\begin{aligned} s(t) &\longrightarrow S(F) \\ s(t - t_0) &\longrightarrow S(F) \cdot \exp(-j2\pi F t_0) \end{aligned}$$

Why Squaring?

Consider an alternating input sequence

$$10101010\cdots \rightarrow +1, -1, +1, -1, +1, -1, +1, -1, \cdots$$

After pulse shaping, the signal essentially becomes like a sinusoid with period $2T_M$

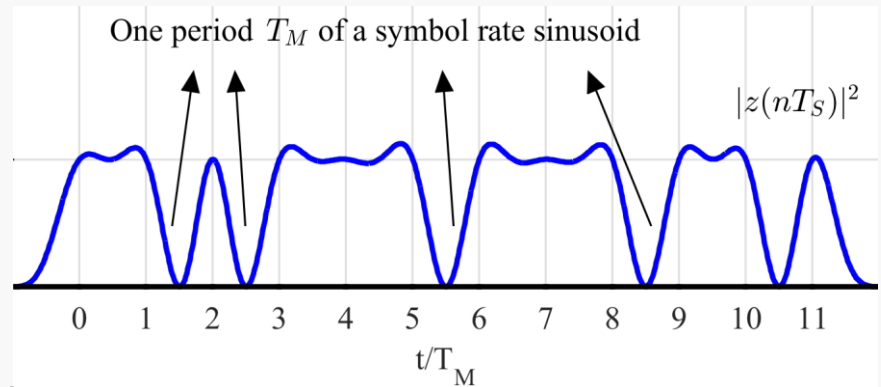


This is the reason why the frequency $1/2T_M$ plays such a central role in the spectrum of all these single-carrier systems!

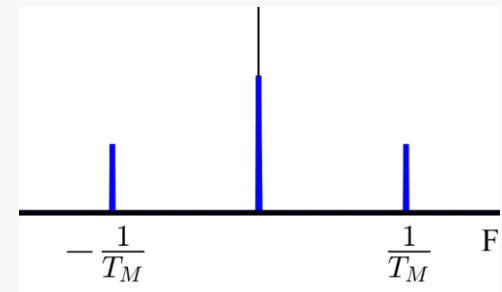
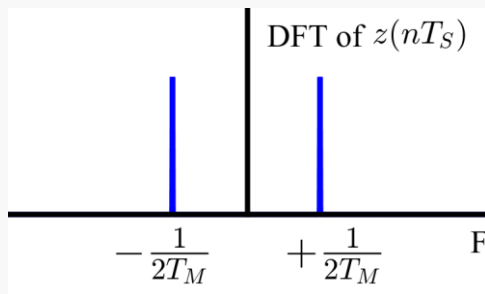
Why Squaring?

What happens if we square this alternating signal?

$$\cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta)$$



Timing lines appear at $\pm 1/T_M$ along with a DC term



Digital Filter and Square Estimator

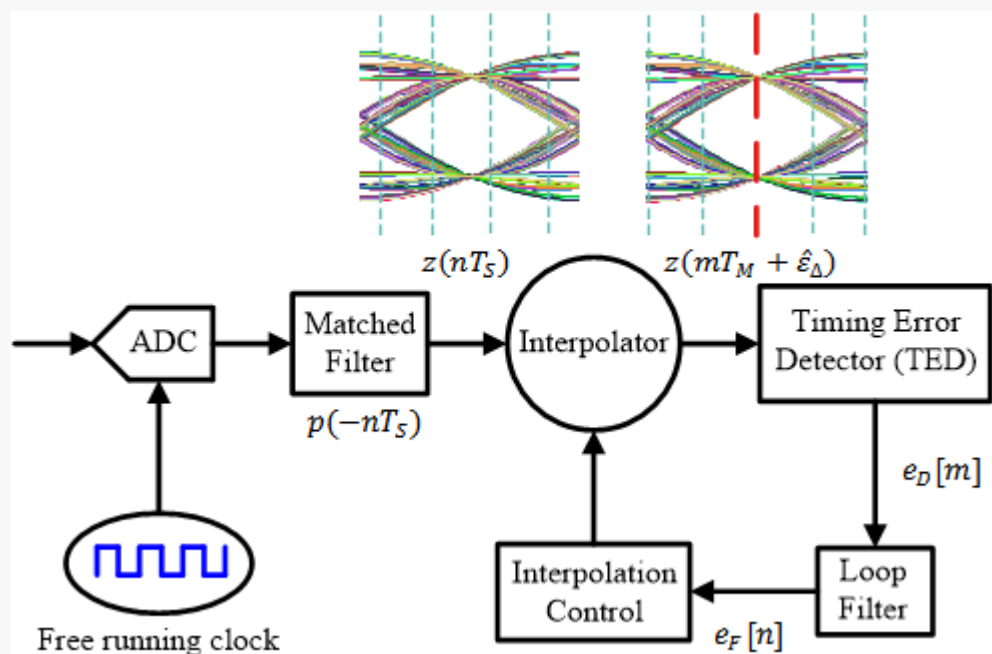
- Since $a[m]$ is unknown, a logical option is to use the matched filter output $z(mT_M + \hat{\varepsilon}_\Delta)$ in its place
- Take the DFT of the output $Y[k] = \text{DFT } |z(nT_S)|^2$
- Recall that timing line is generated at $\pm 1/T_M$
- For N_d symbols,

$$\frac{k}{N} = \pm \frac{1}{L} \rightarrow k = \pm N_d$$

$$\hat{\varepsilon}_\Delta = -\frac{1}{2\pi} \angle Y[N_d]$$

Timing Locked Loop

- Timing Error Detector (TED)
- Loop filter
- **Interpolator:** Computes the 'missing samples' $z(mT_M + \hat{\varepsilon}_\Delta)$
- **Interpolator control:** Provides the basepoint index and the fractional interval



Timing Error Detector (TED)

- Timing synchronization is about balancing like a seesaw
- Refer to this figure often for what follows



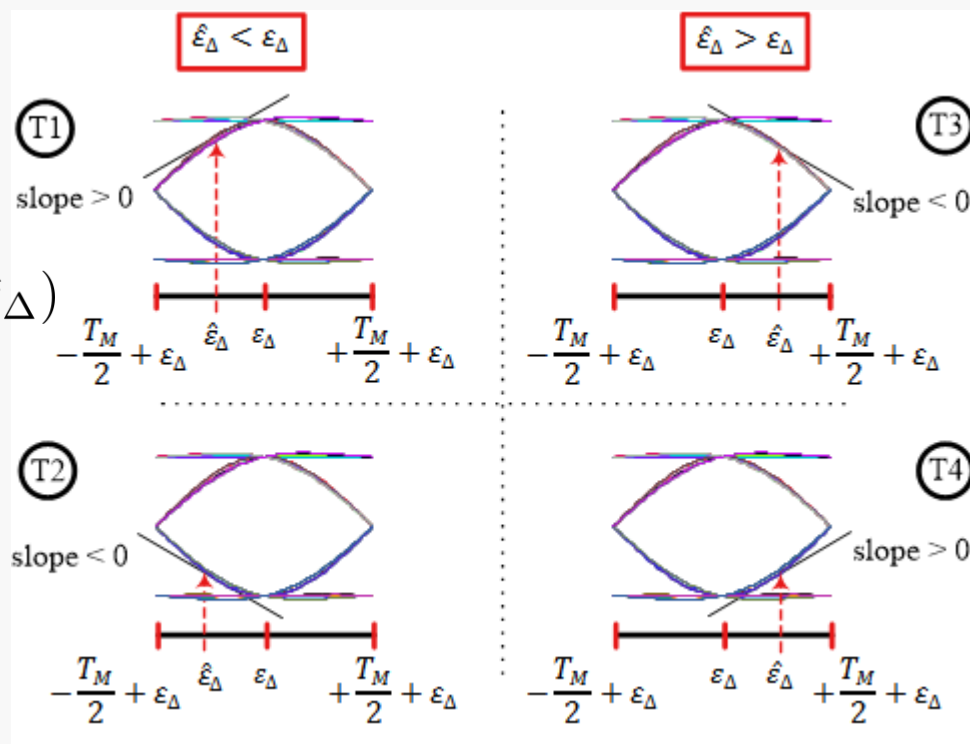
Derivative TED

Derivative is the slope of the curve at a point

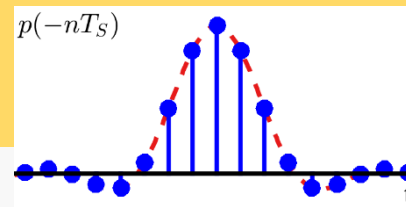
- When the slope reaches zero, the curve is at its maximum or minimum value

Maximize the correlation

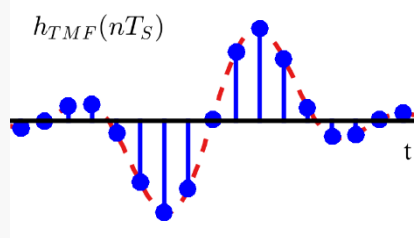
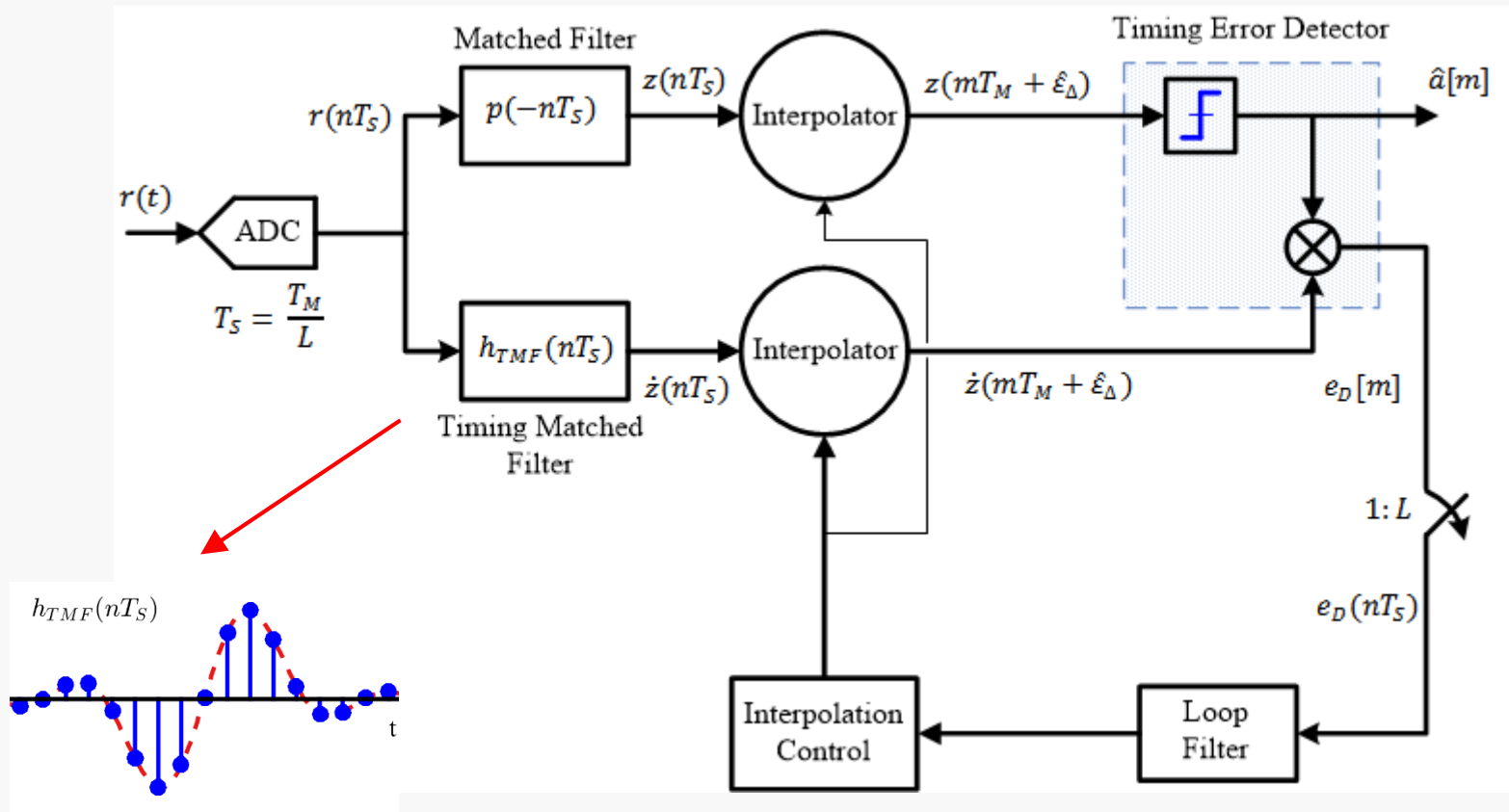
$$e_D[m] = z(mT_M + \hat{\varepsilon}_\Delta) \cdot \dot{z}(mT_M + \hat{\varepsilon}_\Delta)$$



Derivative TED



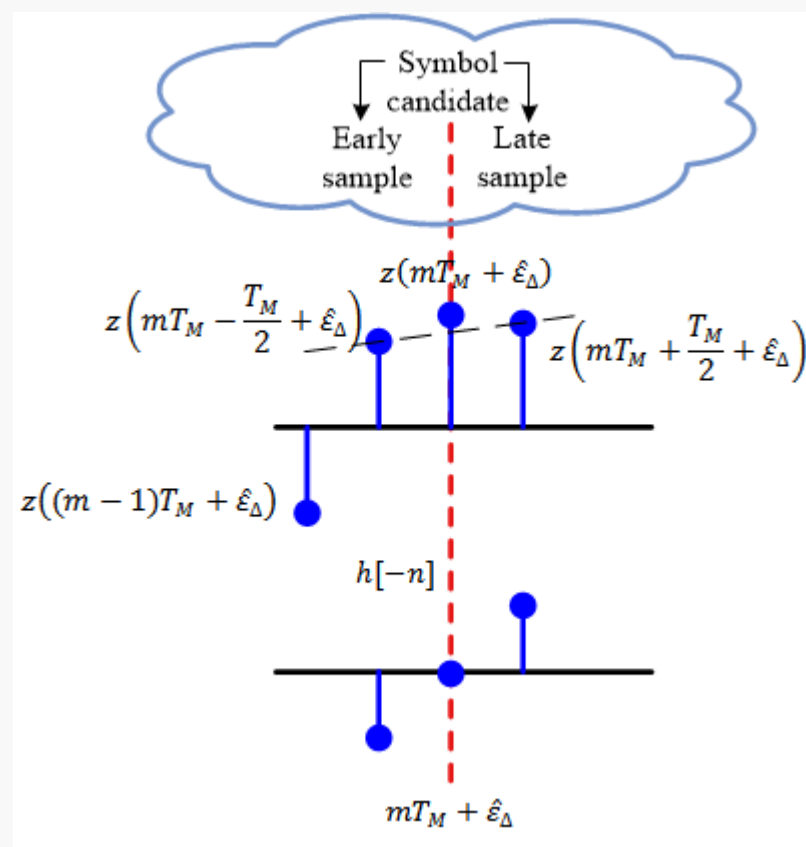
Timing Matched Filter (TMF) (pre-computed derivative of matched filter w.r.t. time)



Early-Late TED

$$e_D[m] = z(mT_M + \hat{\varepsilon}_\Delta) \left\{ z\left(mT_M + \frac{T_M}{2} + \hat{\varepsilon}_\Delta\right) - z\left(mT_M - \frac{T_M}{2} + \hat{\varepsilon}_\Delta\right) \right\}$$

Compute the derivative with the help
of $T_M/2$ -spaced samples



Early-Late TED

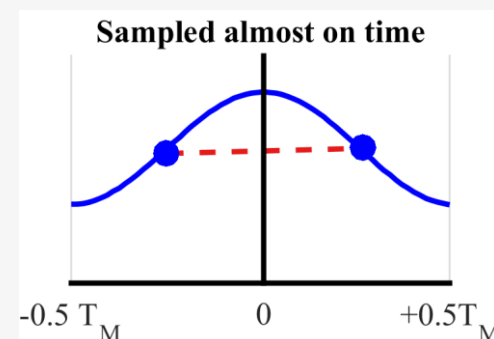
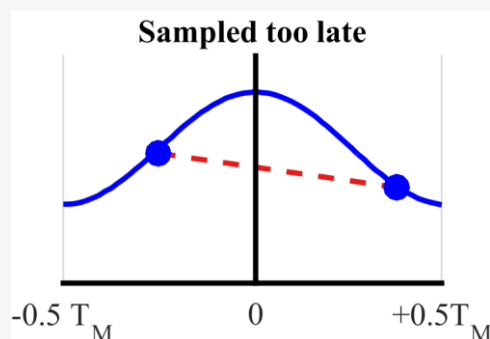
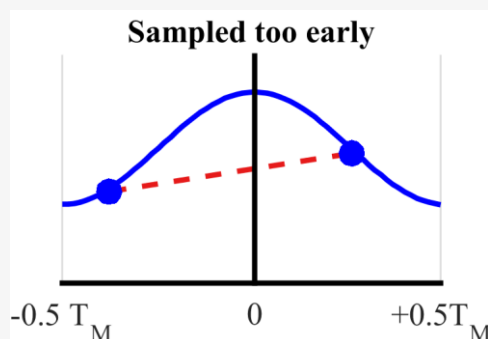
Recall that everything started with $\frac{\partial}{\partial \hat{\epsilon}_\Delta} \left| z(mT_M + \hat{\epsilon}_\Delta) \right|^2$

- Square-law TED

$$e_D[m] = |z(mT_M + \hat{\epsilon}_\Delta + \delta)|^2 - |z(mT_M + \hat{\epsilon}_\Delta - \delta)|^2$$

- Absolute-value TED

$$e_D[m] = |z(mT_M + \hat{\epsilon}_\Delta + \delta)| - |z(mT_M + \hat{\epsilon}_\Delta - \delta)|$$



seesaw

Zero Crossing TED

Symbol centric

$$e_D[m] = z(mT_M + \hat{\varepsilon}_\Delta) \cdot \underbrace{\dot{z}(mT_M + \hat{\varepsilon}_\Delta)}_{\text{drive this term towards zero}}$$

Zero crossing

$$e_D[m] = \underbrace{z(mT_M + \hat{\varepsilon}_\Delta)}_{\text{drive this term towards zero}} \cdot \{ -\dot{z}(mT_M + \hat{\varepsilon}_\Delta) \}$$

Non-data-aided version is known as Gardner TED

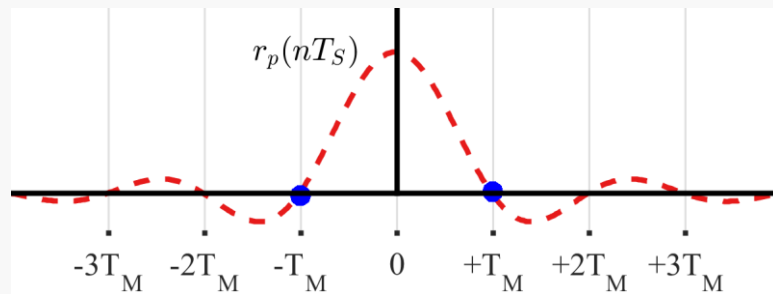
$$e_D[m] = z\left(mT_M - \frac{T_M}{2} + \hat{\varepsilon}_\Delta\right) \left\{ z((m-1)T_M + \hat{\varepsilon}_\Delta) - z(mT_M + \hat{\varepsilon}_\Delta) \right\}$$

Mueller and Muller (M&M) TED

We are interested in a TED that operates at just 1 sample/symbol (why?)

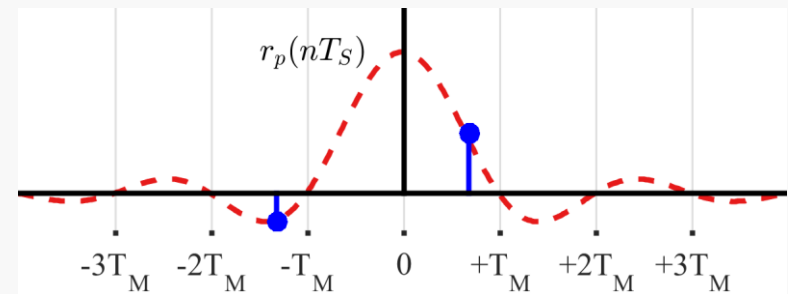
$$\varepsilon_{\Delta} \approx 0$$

$$r_p(+T_M - \varepsilon_{\Delta}) \approx r_p(-T_M - \varepsilon_{\Delta})$$



$$\varepsilon_{\Delta} \neq 0$$

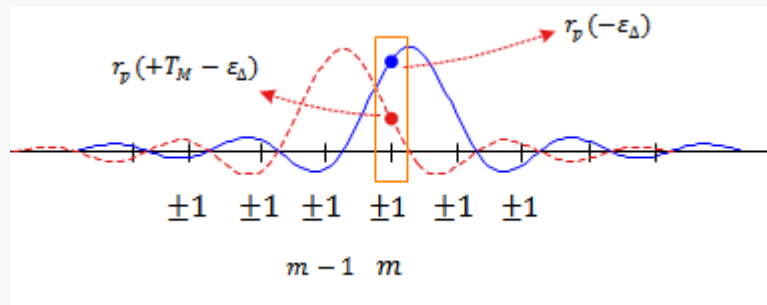
Asymmetry becomes visible



Mueller and Muller (M&M) TED

- Generating $r_p(+T_M - \varepsilon_\Delta)$: Remove $a[m - 1]$ modulation
- Generating $r_p(-T_M - \varepsilon_\Delta)$: Remove $a[m]$ modulation

$$e_D[m] = a[m - 1]z(mT_M + \hat{\varepsilon}_\Delta) - a[m]z((m - 1)T_M + \hat{\varepsilon}_\Delta)$$

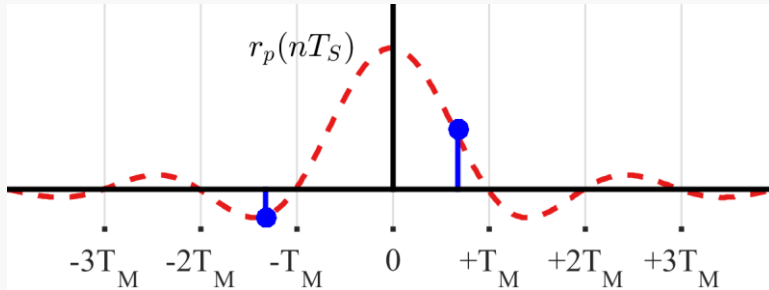


Mueller and Muller (M&M) TED

Why it works better with small excess bandwidth α ?

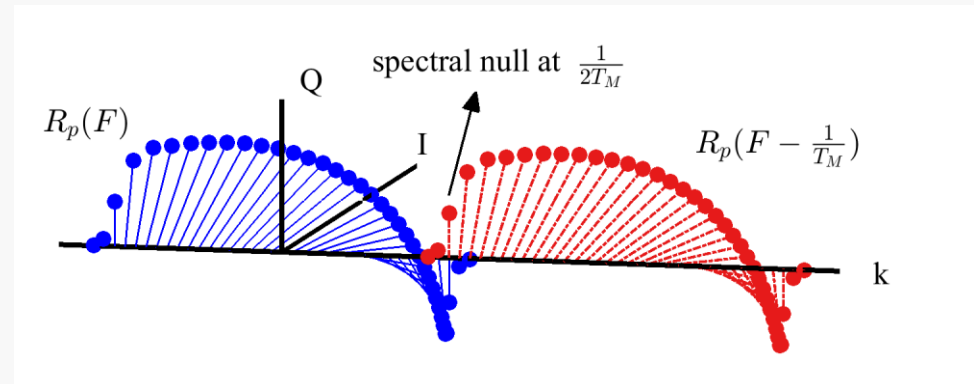
Time domain

Sidelobes provide more timing information



Frequency domain

Less cancellation from the spectral null

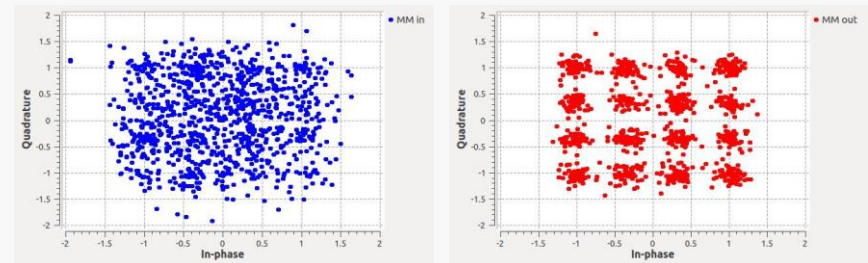


GNU Radio Example

Clock Recovery MM block

- Omega
- Gain omega
- μ
- Gain μ
- Omega relative limit

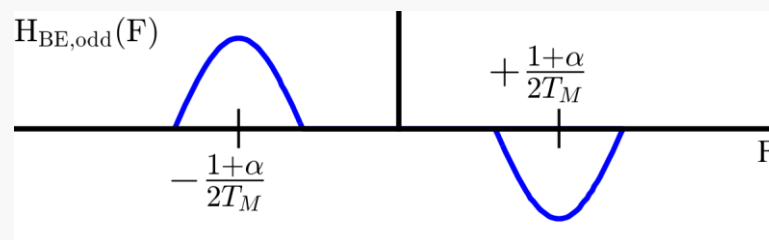
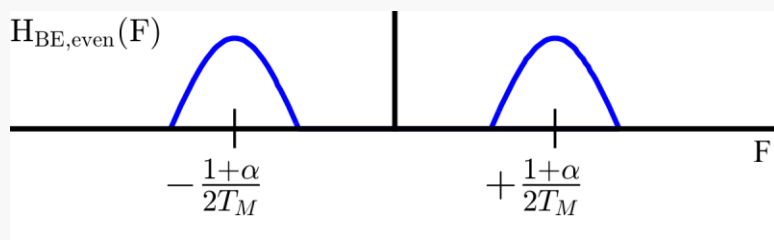
Example: 16-QAM, $\alpha = 0.1$, $L = 4$



Band Edge TED

Review

- Even and odd band edge filters



- Error signal for CFO correction was

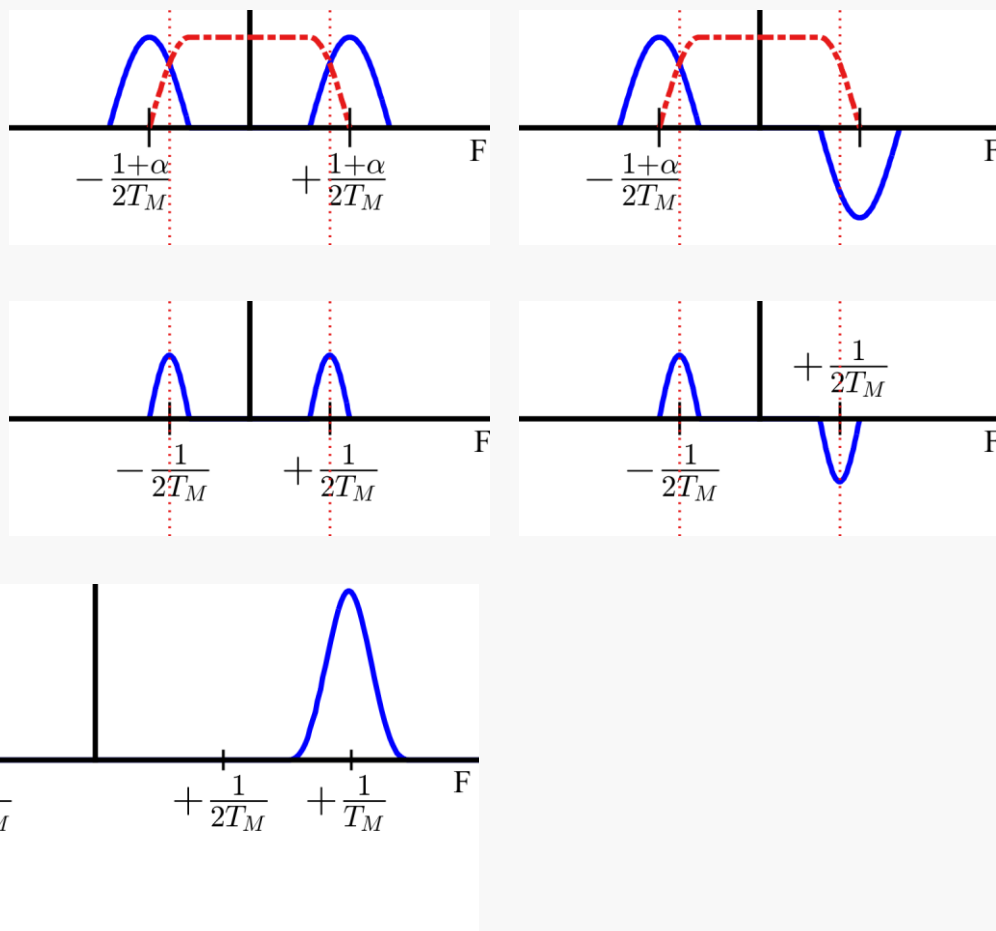
$$e_D[n] = \left\{ z_{BE,even}(nT_S) \cdot z_{BE,odd}^*(nT_S) \right\}_I$$

- Q part of this error signal drives the timing

Band Edge TED

Zero CFO case
$$e_D[n] = \left\{ z_{\text{BE,even}}(nT_S) \cdot z_{\text{BE,odd}}^*(nT_S) \right\}_Q$$

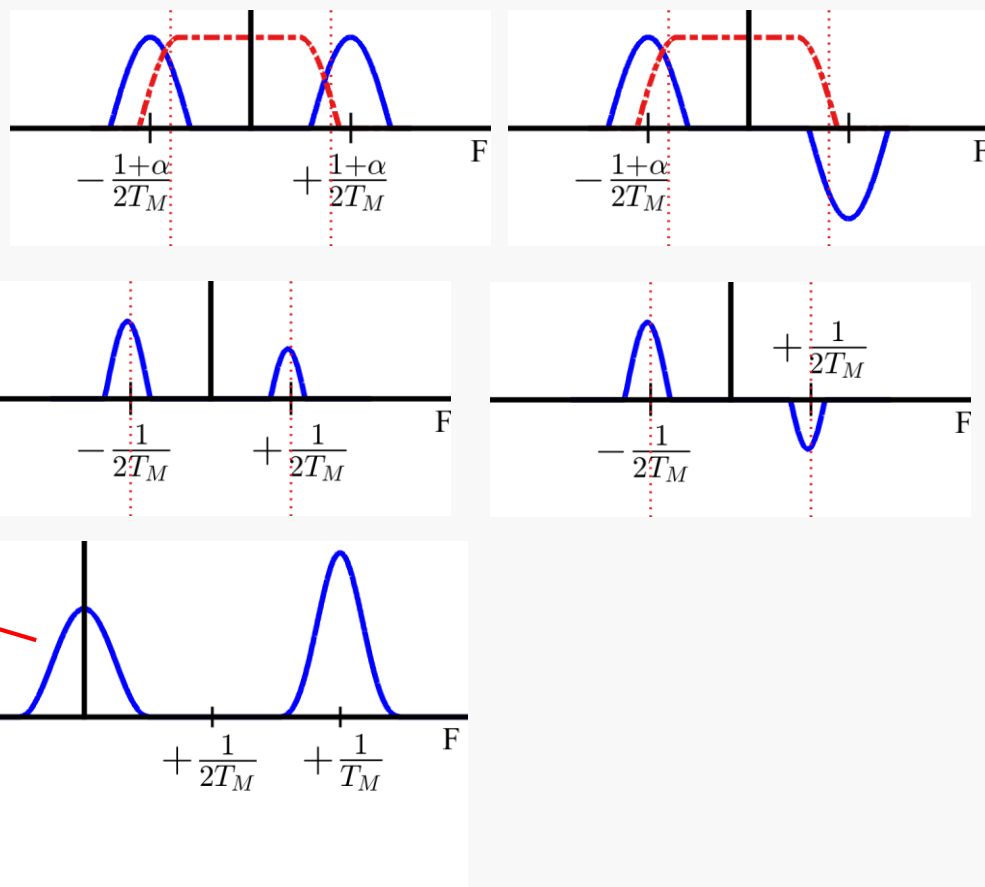
- Multiplication in time is convolution in frequency
- Similar to squaring the signal



Band Edge TED

Non-zero CFO case
$$e_D[n] = \left\{ z_{\text{BE,even}}(nT_S) \cdot z_{\text{BE,odd}}^*(nT_S) \right\}_Q$$

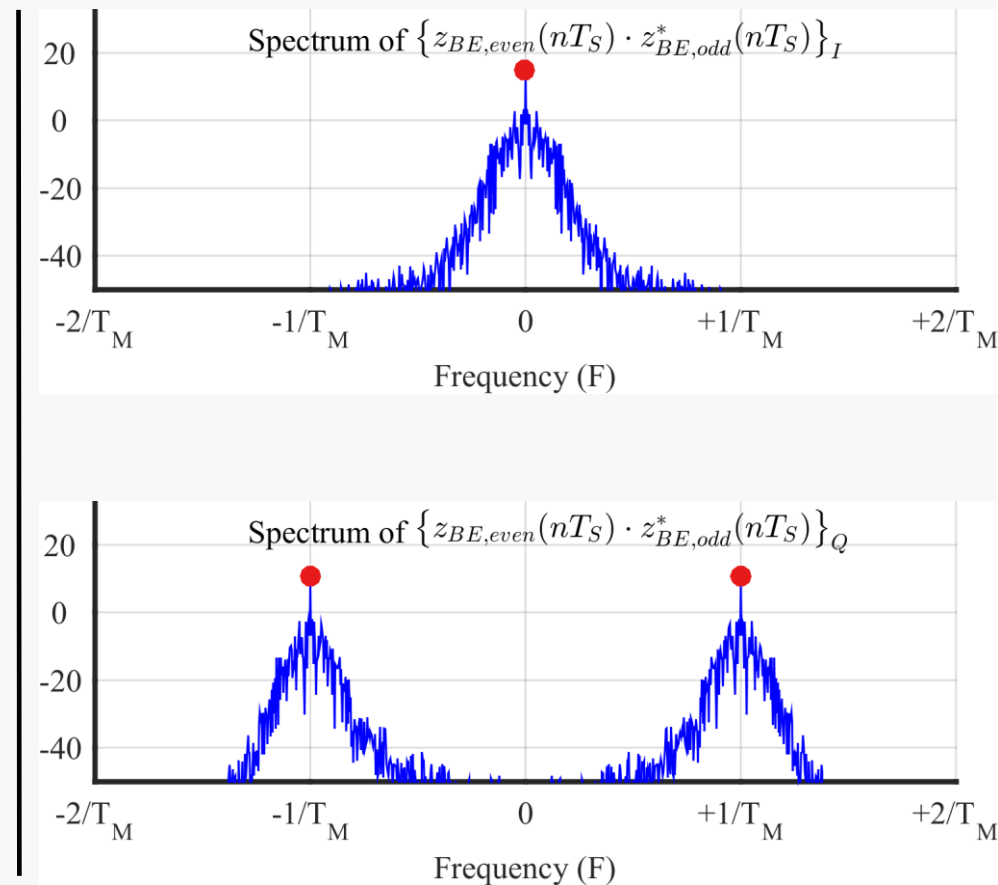
Incoming signal arrives shifted away from the centers of both filters



Band Edge TED

Example with $\alpha = 0.25$ and $L = 4$ samples/symbol

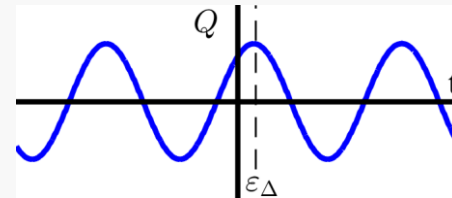
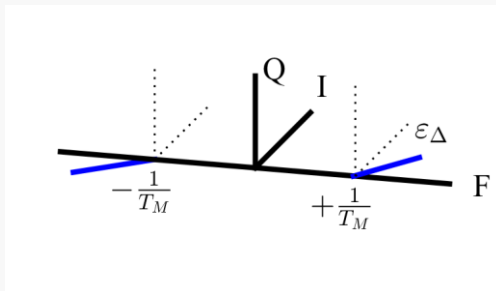
Notice how band edge filters have eliminated much of the self-noise!



Band Edge TED

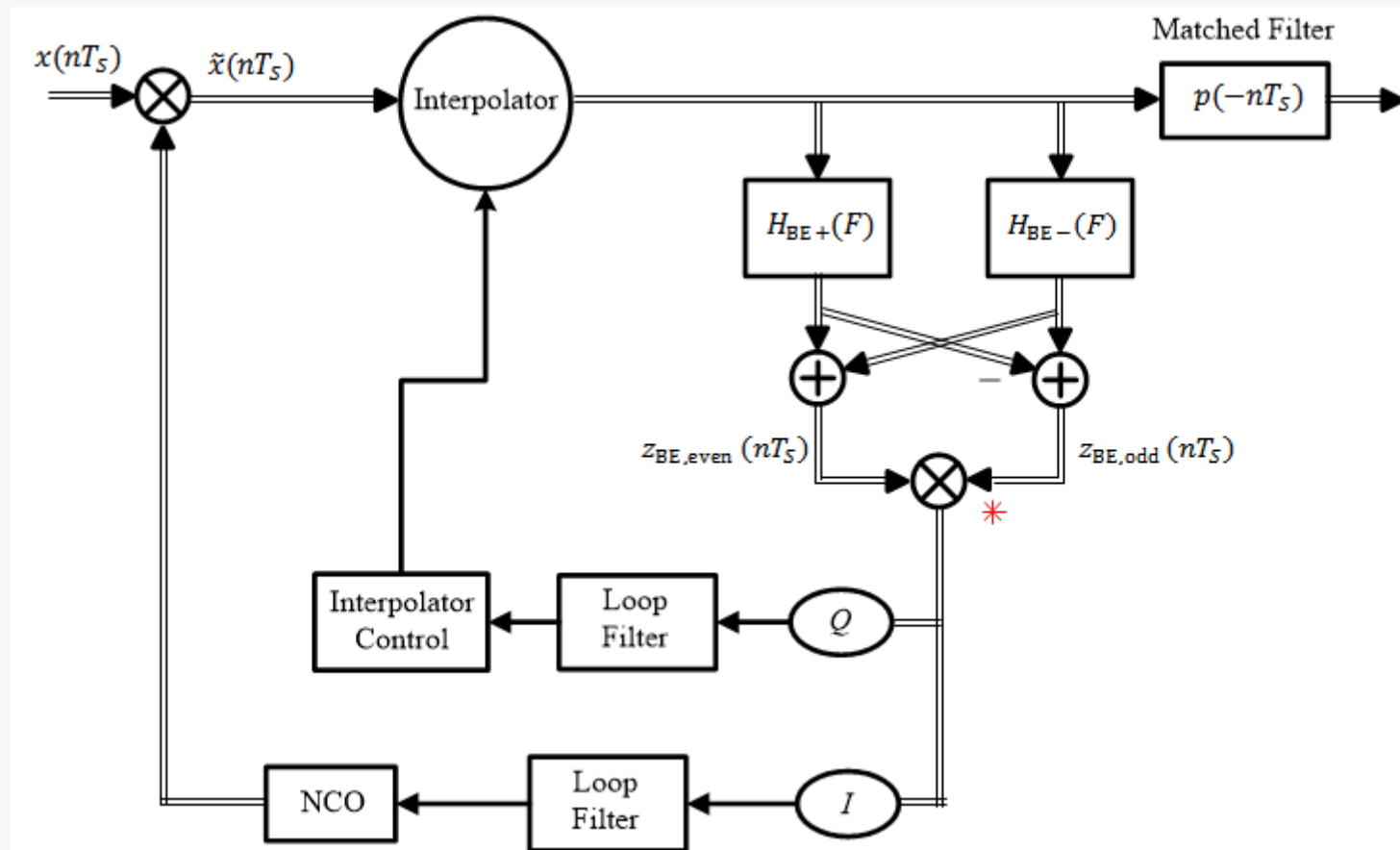
Where is the timing offset ϵ_{Δ} ?

Phase of the above symbol rate sinusoid



A summary of band edge TED

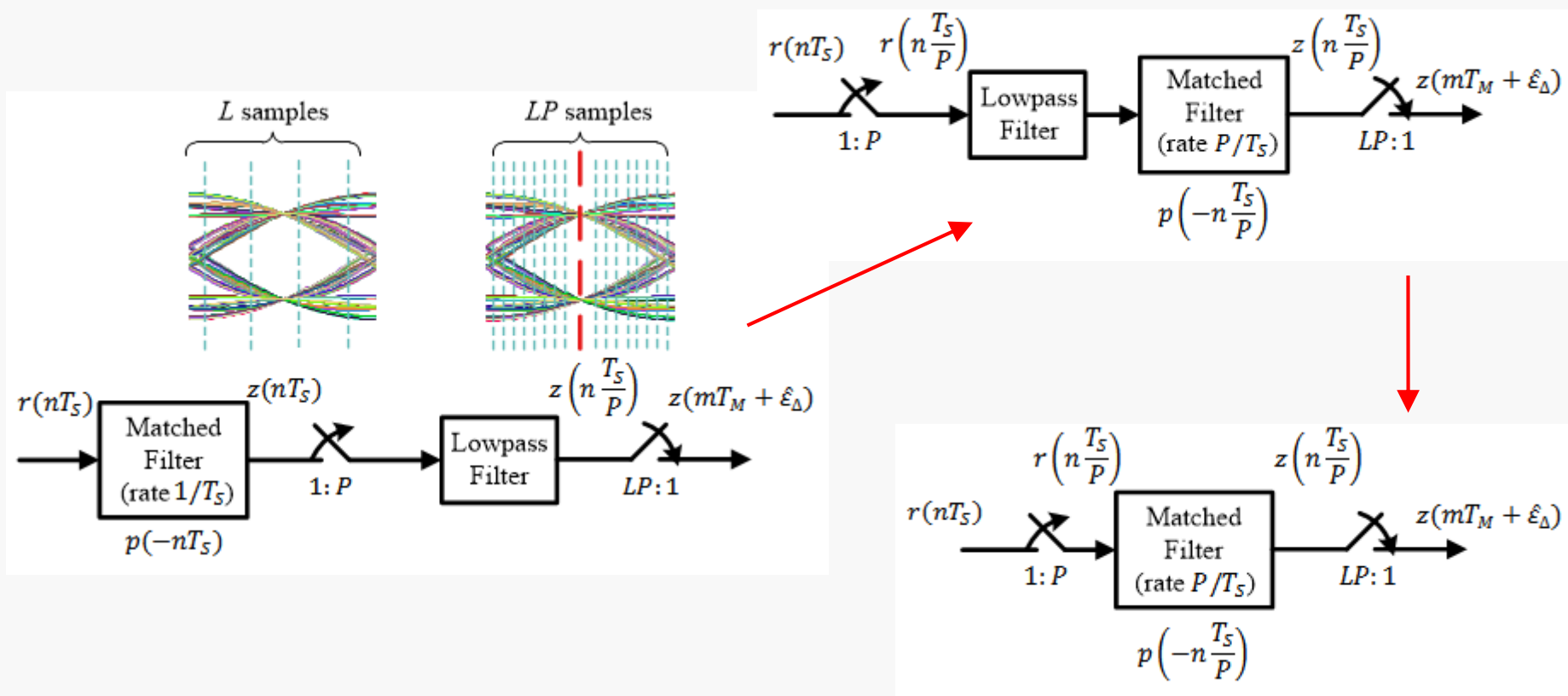
Band Edge TED



Polyphase Clock Synchronizer

Exactly the same as previous timing synchronization methodologies

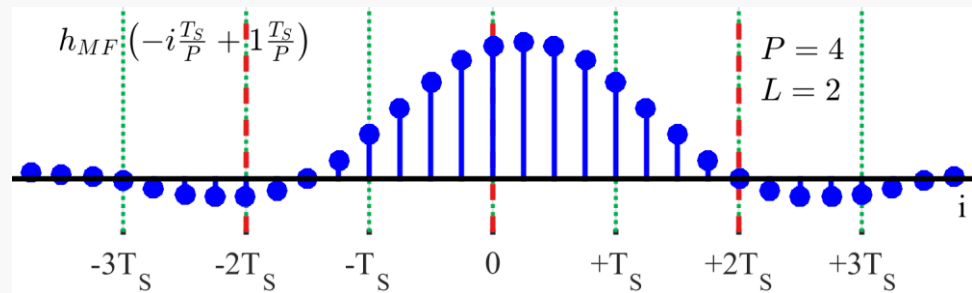
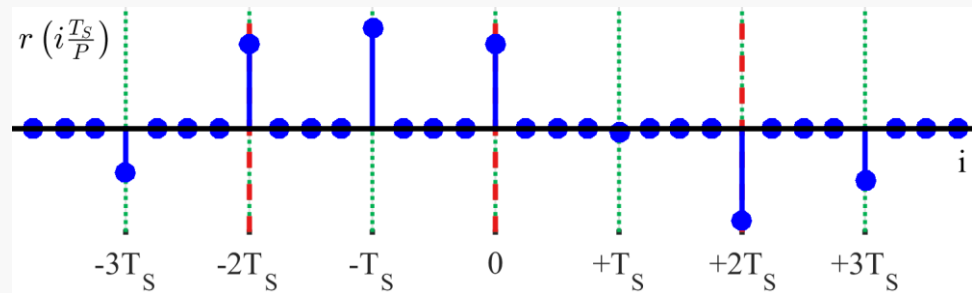
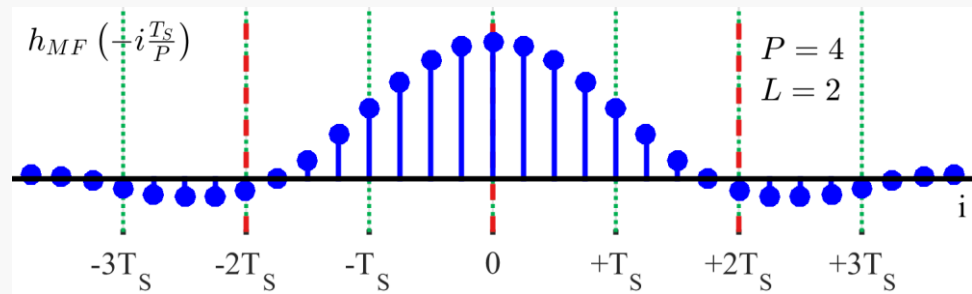
- TED, loop filter, and the interpolator control block remain the same



Polyphase Clock Synchronization

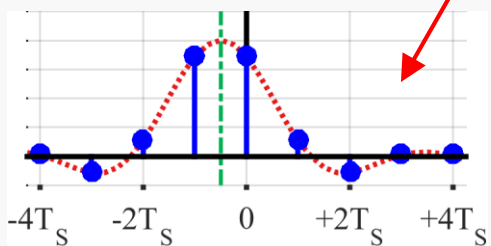
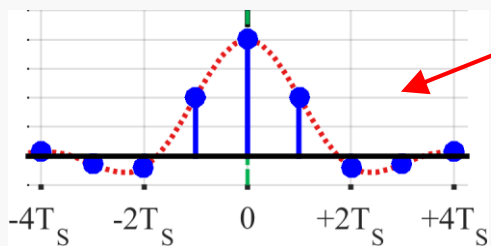
High rate convolution

- $P - 1$ out of P samples in upsampled signal are zero
- Only $1/P$ of the multiplications at each convolution step are required

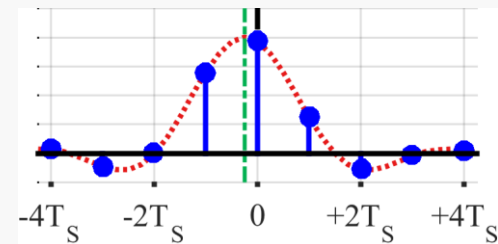
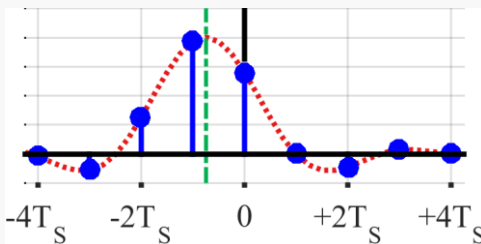


Polyphase Clock Synchronization

Polyphase partition



Polyphase Partition ↓	Filter Samples → → →		
$h_{MF,0}(nT_s) \rightarrow$	$h_{MF} \left[0 \frac{T_s}{P} \right]$	$h_{MF} \left[P \frac{T_s}{P} \right]$	$h_{MF} \left[2P \frac{T_s}{P} \right]$
$h_{MF,1}(nT_s) \rightarrow$	$h_{MF} \left[1 \frac{T_s}{P} \right]$	$h_{MF} \left[(P+1) \frac{T_s}{P} \right]$	$h_{MF} \left[(2P+1) \frac{T_s}{P} \right]$
$\vdots$	$\dots$	$\dots$	$\dots$
$h_{MF,\delta}(nT_s) \rightarrow$	$h_{MF} \left[\delta \frac{T_s}{P} \right]$	$h_{MF} \left[(P+\delta) \frac{T_s}{P} \right]$	$h_{MF} \left[(2P+\delta) \frac{T_s}{P} \right]$
$\vdots$	$\dots$	$\dots$	$\dots$
$h_{MF,P-1}(nT_s) \rightarrow$	$h_{MF} \left[(P-1) \frac{T_s}{P} \right]$	$h_{MF} \left[(2P-1) \frac{T_s}{P} \right]$	$h_{MF} \left[(3P-1) \frac{T_s}{P} \right]$

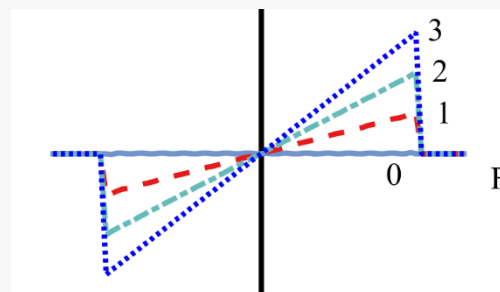
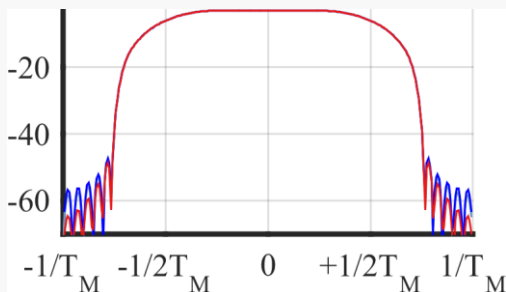
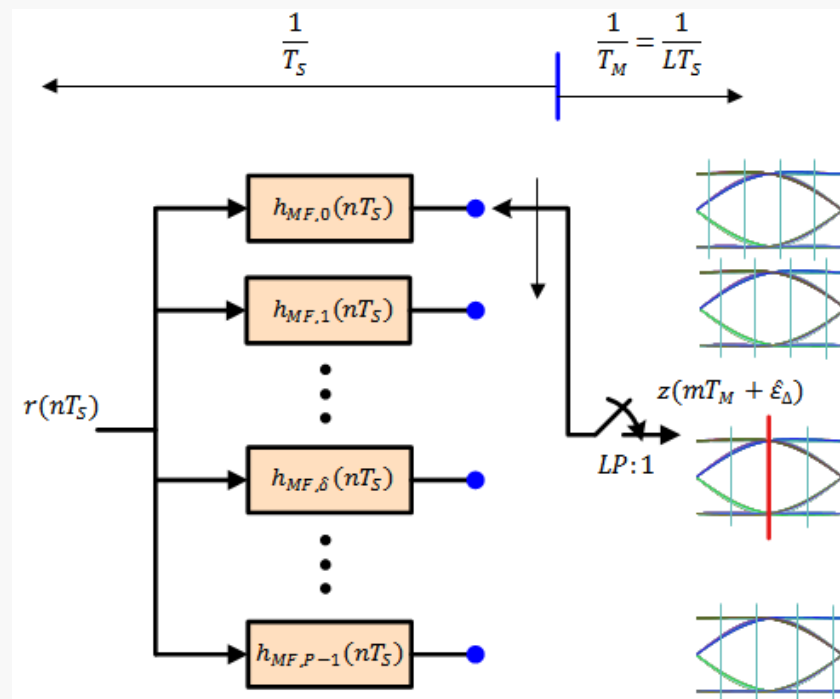


Polyphase Clock Synchronizer

- The ratio δ/P plays the same role as the fractional interval μ_m in the interpolation filter

$$\mu_m = \frac{\delta}{P}$$

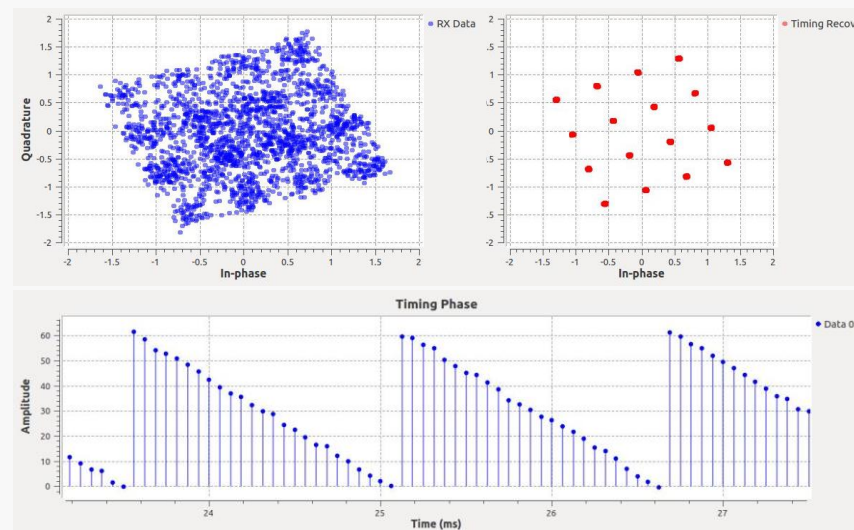
- Where is the TED?



GNU Radio Example

Polyphase Clock Sync block

- Samples/symbol
- Loop bandwidth
- Taps
- Filter size
- Initial phase
- Maximum rate deviation



Example: 16-QAM, loop bandwidth $2\pi/150$,

$$L = 4, \alpha = 0.5, P = 64$$

Small Picture

Energy maximization perspective

- Spectral line generation through squaring the signal and track the sinusoid through a PLL
- Explicitly search for this peak using the derivative, zero crossing or band edge approach